

Quiz # 2 Solution

Course: CSC354 – Machine Learning

Semester: 7th (FA22-BCE-A/B)

Total Marks = 15

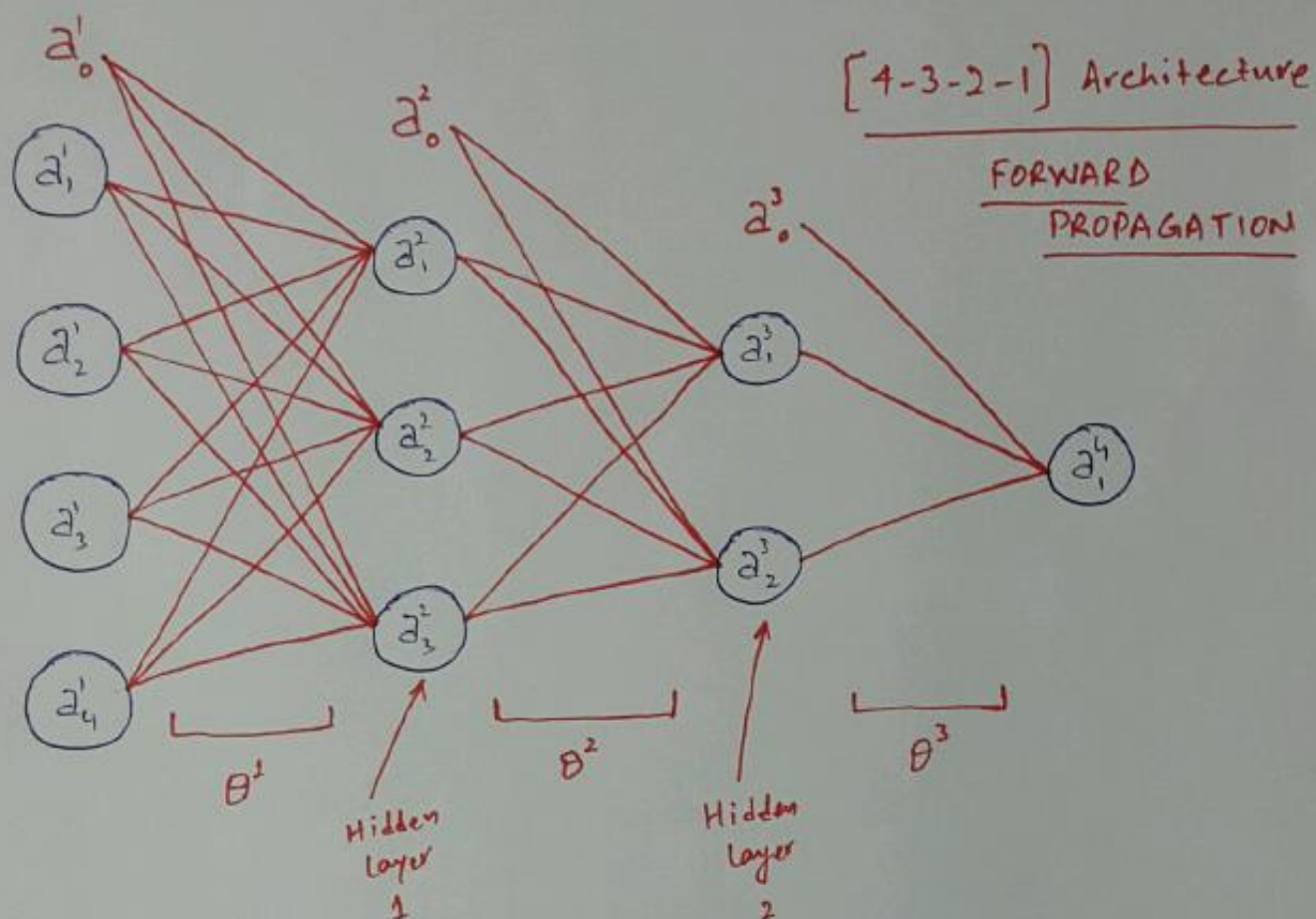
Date: 24-10-2025

Time Allowed: 20 Min

Question 1: Relate the concepts of artificial neural network to write the mathematical expressions for network nodes (neurons) during forward and backward propagation.

(CLO-2, PLO-2, C3) (15 Marks)

Assume an artificial neural network with an architecture of 4-3-2-1, i.e., 4 input features, 1 output unit, two hidden layers with 3 and 2 units in hidden layer 1 and 2 respectively (without the bias unit). Illustrate the complete network architecture with connections between layers. Produce the mathematical expression for each activation units ($\mathbf{a}_s$) via forward propagation, and the error in each units (δ_s) via backward propagation. (Add bias units while forward propagation).



Hidden Layer 1

$$\begin{cases} a''_1 = g \left(\theta_{10}^{(1)} a'_0 + \theta_{11}^{(1)} a'_1 + \theta_{12}^{(1)} a'_2 + \theta_{13}^{(1)} a'_3 + \theta_{14}^{(1)} a'_4 \right) \\ a''_2 = g \left(\theta_{20}^{(1)} a'_0 + \theta_{21}^{(1)} a'_1 + \theta_{22}^{(1)} a'_2 + \theta_{23}^{(1)} a'_3 + \theta_{24}^{(1)} a'_4 \right) \\ a''_3 = g \left(\theta_{30}^{(1)} a'_0 + \theta_{31}^{(1)} a'_1 + \theta_{32}^{(1)} a'_2 + \theta_{33}^{(1)} a'_3 + \theta_{34}^{(1)} a'_4 \right) \end{cases}$$

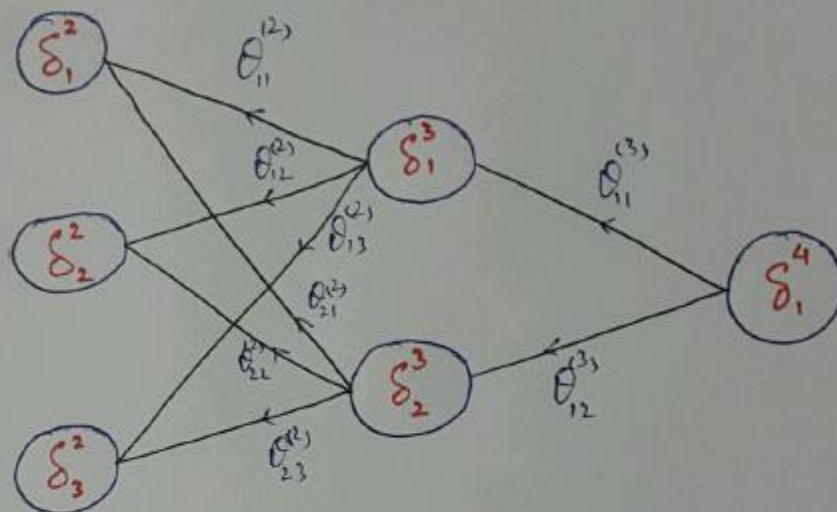
Hidden Layer 2

$$\begin{cases} a'''_1 = g \left(\theta_{10}^{(2)} a''_0 + \theta_{11}^{(2)} a''_1 + \theta_{12}^{(2)} a''_2 + \theta_{13}^{(2)} a''_3 \right) \\ a'''_2 = g \left(\theta_{20}^{(2)} a''_0 + \theta_{21}^{(2)} a''_1 + \theta_{22}^{(2)} a''_2 + \theta_{23}^{(2)} a''_3 \right) \end{cases}$$

Output

$$a''''_1 = g \left(\theta_{10}^{(3)} a'''_0 + \theta_{11}^{(3)} a'''_1 + \theta_{12}^{(3)} a'''_2 \right)$$

BACKWARD PROPAGATION



Output Error $\left[\delta_1^{(4)} = a_1^4 - y \right]$

Error at Hidden Layer 2 $\left\{ \begin{aligned} \delta_1^{(3)} &= \theta_{11}^{(3)} \delta_1^{(4)} + \cancel{\theta_{12}^{(3)} \delta_1^{(4)}} \\ \delta_2^{(3)} &= \theta_{12}^{(3)} \delta_1^{(4)} \end{aligned} \right.$

Error at Hidden Layer 1 $\left\{ \begin{aligned} \delta_1^{(2)} &= \theta_{11}^{(2)} \delta_1^{(3)} + \theta_{22}^{(2)} \delta_2^{(3)} + \cancel{\theta_{13}^{(2)} \delta_1^{(3)}} \\ \delta_2^{(2)} &= \theta_{12}^{(2)} \delta_1^{(3)} + \theta_{22}^{(2)} \delta_2^{(3)} \\ \delta_3^{(2)} &= \theta_{13}^{(2)} \delta_1^{(3)} + \theta_{23}^{(2)} \delta_2^{(3)} \end{aligned} \right.$